

Labsheet- 9

Data Storytelling using Superstore Dataset

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In [ ]: import pandas as pd
import plotly.express as px

# Step 1: Load Dataset
df = pd.read_csv("Sample - Superstore.csv", encoding="latin1")
print("Shape:", df.shape)
print(df.head())

# Step 2: Check Missing Values
print("\nMissing Values:")
print(df.isnull().sum())

# Step 3: Convert Date
df["Order Date"] = pd.to_datetime(df["Order Date"], format="mixed")
df["Year"] = df["Order Date"].dt.year

# Step 4: KPIs
print("\nKPIs")
print("Total Sales:", round(df["Sales"].sum(), 2))
print("Total Profit:", round(df["Profit"].sum(), 2))
print("Total Orders:", df["Order ID"].nunique())

# Step 5: Sales by Category
data = df.groupby("Category", as_index=False)["Sales"].sum()
px.bar(data, x="Category", y="Sales",
        title="Sales by Category").show()

# Step 6: Sales by Region
data = df.groupby("Region", as_index=False)["Sales"].sum()
px.bar(data, x="Region", y="Sales",
        title="Sales by Region").show()

# Step 7: Yearly Sales
data = df.groupby("Year", as_index=False)["Sales"].sum()
px.line(data, x="Year", y="Sales",
         markers=True, title="Yearly Sales Trend").show()

# Step 8: Profit by Category
data = df.groupby("Category", as_index=False)["Profit"].sum()
px.bar(data, x="Category", y="Profit",
        title="Profit by Category").show()

# Step 9: Insights
print("\nBusiness Insights:")
print("1. Sales vary across categories and regions.")
print("2. Yearly sales show business trends.")
print("3. High sales do not always mean high profit.")
print("4. Profit analysis identifies profitable categories.")

# Step 10: Recommendations
print("\nRecommendations:")
print("1. Focus on high-performing categories.")
print("2. Improve low-performing regions.")
```

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print("3. Control discounts to maintain profit.")  
print("4. Plan inventory using sales trends.")
```

Shape: (9994, 21)

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	\
0	1	CA-2016-152156	11-08-2016	11-11-2016	Second Class	CG-12520	
1	2	CA-2016-152156	11-08-2016	11-11-2016	Second Class	CG-12520	
2	3	CA-2016-138688	06-12-2016	6/16/2016	Second Class	DV-13045	
3	4	US-2015-108966	10-11-2015	10/18/2015	Standard Class	SO-20335	
4	5	US-2015-108966	10-11-2015	10/18/2015	Standard Class	SO-20335	

	Customer Name	Segment	Country	City	...	\
0	Claire Gute	Consumer	United States	Henderson	...	
1	Claire Gute	Consumer	United States	Henderson	...	
2	Darrin Van Huff	Corporate	United States	Los Angeles	...	
3	Sean O'Donnell	Consumer	United States	Fort Lauderdale	...	
4	Sean O'Donnell	Consumer	United States	Fort Lauderdale	...	

	Postal Code	Region	Product ID	Category	Sub-Category	\
0	42420	South	FUR-BO-10001798	Furniture	Bookcases	
1	42420	South	FUR-CH-10000454	Furniture	Chairs	
2	90036	West	OFF-LA-10000240	Office Supplies	Labels	
3	33311	South	FUR-TA-10000577	Furniture	Tables	
4	33311	South	OFF-ST-10000760	Office Supplies	Storage	

	Product Name	Sales	Quantity	\
0	Bush Somerset Collection Bookcase	261.9600	2	
1	Hon Deluxe Fabric Upholstered Stacking Chairs,...	731.9400	3	
2	Self-Adhesive Address Labels for Typewriters b...	14.6200	2	
3	Bretford CR4500 Series Slim Rectangular Table	957.5775	5	
4	Eldon Fold 'N Roll Cart System	22.3680	2	

	Discount	Profit
0	0.00	41.9136
1	0.00	219.5820
2	0.00	6.8714
3	0.45	-383.0310
4	0.20	2.5164

[5 rows x 21 columns]

Missing Values:

Row ID	0
Order ID	0
Order Date	0
Ship Date	0
Ship Mode	0
Customer ID	0
Customer Name	0
Segment	0
Country	0
City	0
State	0
Postal Code	0
Region	0
Product ID	0
Category	0
Sub-Category	0
Product Name	0
Sales	0
Quantity	0
Discount	0
Profit	0

dtype: int64

KPIs

Total Sales: 2297200.86

Total Profit: 286397.02

Total Orders: 5009

Business Insights:

1. Sales vary across categories and regions.
2. Yearly sales show business trends.
3. High sales do not always mean high profit.
4. Profit analysis identifies profitable categories.

Recommendations:

1. Focus on high-performing categories.
2. Improve low-performing regions.
3. Control discounts to maintain profit.
4. Plan inventory using sales trends.

Questions and Answers

1. What is data storytelling? How is it different from data visualization?

Data storytelling is the process of combining **data, visualizations, and narrative** to communicate meaningful insights. Data visualization mainly presents data through charts and graphs, while data storytelling explains the **meaning, importance, and context** behind the visualized data.

2. Why is storytelling important in business analytics and decision-making?

Storytelling makes complex analytical information easier to understand. It helps business stakeholders identify important trends, understand problems, and make **data-driven decisions** more effectively.

3. What are the essential components of an effective data story?

The main components are **data, visualization, and narrative**. A good data story should also have a clear objective, logical flow, important insights, and actionable recommendations.

4. How do Key Performance Indicators (KPIs) enhance business reporting?

KPIs provide measurable values that show the performance of a business. They help organizations **track progress, identify problems, compare performance, and make informed decisions**.

5. Why should visualizations be arranged in a logical sequence while presenting analytical findings?

A logical sequence helps the audience understand the story step by step. It connects the **problem, analysis, findings, and recommendations**, making the presentation clear and easy to follow.

6. What factors should be considered while selecting visualizations for a business presentation?

The visualization should match the type of data and purpose of the analysis. Factors include **data type, audience, simplicity, readability, comparison requirements, trends, and the message** that needs to be communicated.

7. Explain how dashboards and storytelling complement each other in Business Intelligence.

Dashboards provide interactive views of important **KPIs, trends, and metrics**, while storytelling explains what those results mean. Together, they help stakeholders explore data and understand the insights needed for business decisions.

8. What challenges may arise while communicating analytical insights to non-technical stakeholders?

Common challenges include **technical terminology, complex charts, too much information, lack of context, and different levels of data knowledge**. Using simple language, clear visuals, and relevant examples can overcome these challenges.

9. Give two real-world examples where data storytelling has influenced business or policy decisions.

- **Business:** Companies use sales dashboards and customer analytics to identify changing customer demand and adjust **marketing, inventory, and pricing strategies**.
- **Public policy:** Governments and public-health organizations use COVID-19 case, hospitalization, and vaccination data with visual dashboards to communicate trends and support **health-policy planning**.

10. How can effective data storytelling improve strategic planning and organizational performance?

Effective data storytelling helps organizations understand **performance trends, opportunities, risks, and problems**. It supports better strategic planning, improves communication between teams, and helps managers take actions based on evidence, which can improve organizational performance.